Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 03)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Give **three** sources of unburnt hydrocarbons in urban air.

[3 marks]

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0 1 . 2 Describe **two** methods used to control hydrocarbon emissions at petrol filling stations.

[2 marks]

1 .....

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0 1 . 3 Explain how a catalytic converter controls the emissions of carbon monoxide, hydrocarbons and NOx.

[3 marks]

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0 1 . 4 Explain why carbon monoxide is toxic to humans.

[2 marks]

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0 2 An aquifer beneath an agricultural plain holds 3.6×10^{11} m^3 of water. It is recharged at 1.2×10^9 m^3 per year, and 4.8×10^9 m^3 per year is abstracted from it.

0 2 . 1 Residence time = volume of the reservoir / rate of flow through it. Calculate the residence time of the water in this aquifer, using the recharge rate.

[2 marks]

Answer _____ years

0 2 . 2 Explain the properties a rock must have to form a useful aquifer.

[3 marks]

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0 2 . 3 Explain why the water table beneath the plain is falling.

[2 marks]

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0 2 . 4 Describe **four** consequences of continuing to over-abstract this aquifer.

[4 marks]

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0 2 . 5 Explain why an aquifer holding 'fossil water' cannot be exploited sustainably, and describe **two** measures that would reduce abstraction.

[4 marks]

15

0 3 Figure 1 lists the stages in a water treatment works supplying drinking water from a lowland river.

Figure 1

Stage	Process
1	Screens
2	Sedimentation
3	Aeration
4	Flocculation and clarification
5	Filtration through sand and gravel
6	Activated carbon
7	Sterilisation
8	pH adjustment

0 3 . 1 Explain why clay particles do not settle out in the sedimentation stage, and describe how they are removed in stage 4.

[3 marks]

0 3 . 2 Explain the purpose of aeration (stage 3).

[2 marks]

0 3 . 3 Explain why ozone or UV light may be used for sterilisation instead of chlorine when water comes from peaty upland sources.

[2 marks]

0 3 . 4 Describe how each of the following contaminants is removed from water.

[3 marks]

Toxic metal ions (lead, mercury, arsenic)

Salt

Odours and pesticides

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0 4 . 1 Explain the roles of **three** named groups of soil organisms in maintaining soil fertility.

[6 marks]

0 4 . 2 Explain why a soil pH between 5.5 and 7.0 is best for fertility.

[2 marks]

0 4 . 3 Explain how soil compaction by heavy machinery reduces fertility.

[2 marks]

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0 5 Table 1 shows three specialist solid wastes.

0 5 . 1 Complete Table 1 to show the main hazard of each waste and the disposal method used.

[6 marks]

Table 1

Waste	Main hazard	Disposal method	
Asbestos			
Cyanide compounds			
High-level radioactive waste			

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0 5 . 2 Explain why encapsulated asbestos in cement roof panels is not considered dangerous while it is intact.

[1 mark]

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0 5 . 3 Describe the process of encapsulation used for intermediate-level radioactive waste and wastes containing heavy metals.

[2 marks]

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0 5 . 4 Explain why vitrification is preferred to encapsulation in cement for high-level waste.

[1 mark]

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0 6 . 1 Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

A Ozone is destroyed only by CFCs ☐

B Ozone absorbs mainly infrared radiation ☐

C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐

D Ozone concentrations are highest at ground level ☐

0 6 . 2 Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

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0 6 . 3 Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

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0 7 . 1 Explain the difference between a negative and a positive feedback mechanism.

[2 marks]

0 7 . 2 Describe **two** negative feedback mechanisms that may reduce global warming.

[4 marks]

0 7 . 3 Explain how melting permafrost and melting methane hydrate act as positive feedback mechanisms.

[4 marks]

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0 7 . 4 Give **three** other positive feedback mechanisms.

[3 marks]

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0 7 . 5 Explain what is meant by a climate **tipping point**.

[2 marks]

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0 8 . 1 Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

A Sulfur dioxide from a coal-fired power station ☐

B Tropospheric ozone ☐

C Carbon monoxide from a car exhaust ☐

D Nitrogen monoxide from a diesel engine ☐

0 8 . 2 Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

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0 9 . 1 A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

A The readings are precise but may not be accurate ☐

B The readings are accurate but not precise ☐

C The readings are both accurate and precise ☐

D The readings are reliable because the meter is electronic ☐

0 9 . 2 Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

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0 9 . 3 Explain why a preliminary study is carried out before the main investigation.

[2 marks]

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1 0 . 1 Evaluate the extent to which new technologies could allow fossil fuels and nuclear fission to continue to supply a large proportion of the world's energy sustainably.

[9 marks]

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Extra space

1 0 . 2 State the approximate percentage of natural gas in a reservoir that can be recovered using its own pressure.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

[25 marks]

OR

1 1 . 2 Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS